

TIBS

Theme 2: After Innovation

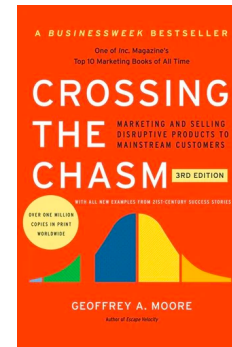
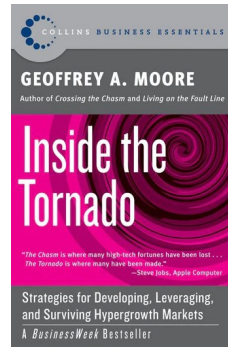
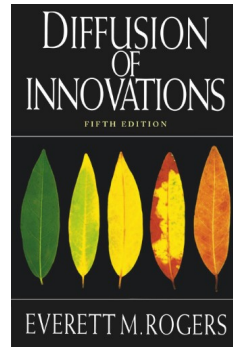
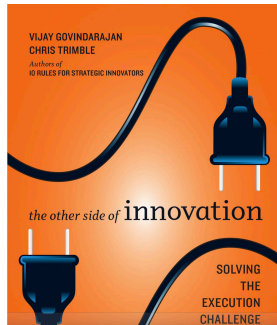
- After you've come up with an innovation, what next? How do innovations spread and become popular?

Overview

- After you've come up with an innovation, what next?
- In commercial contexts, then there is very often an expectation that it should be commercialised in some sense, to generate profit, beyond the R&D and commercialisation costs.
- This process of execution can benefit from an understanding of how innovative ideas, technologies, or products spread through a population of potential users, of potential consumers.

Books

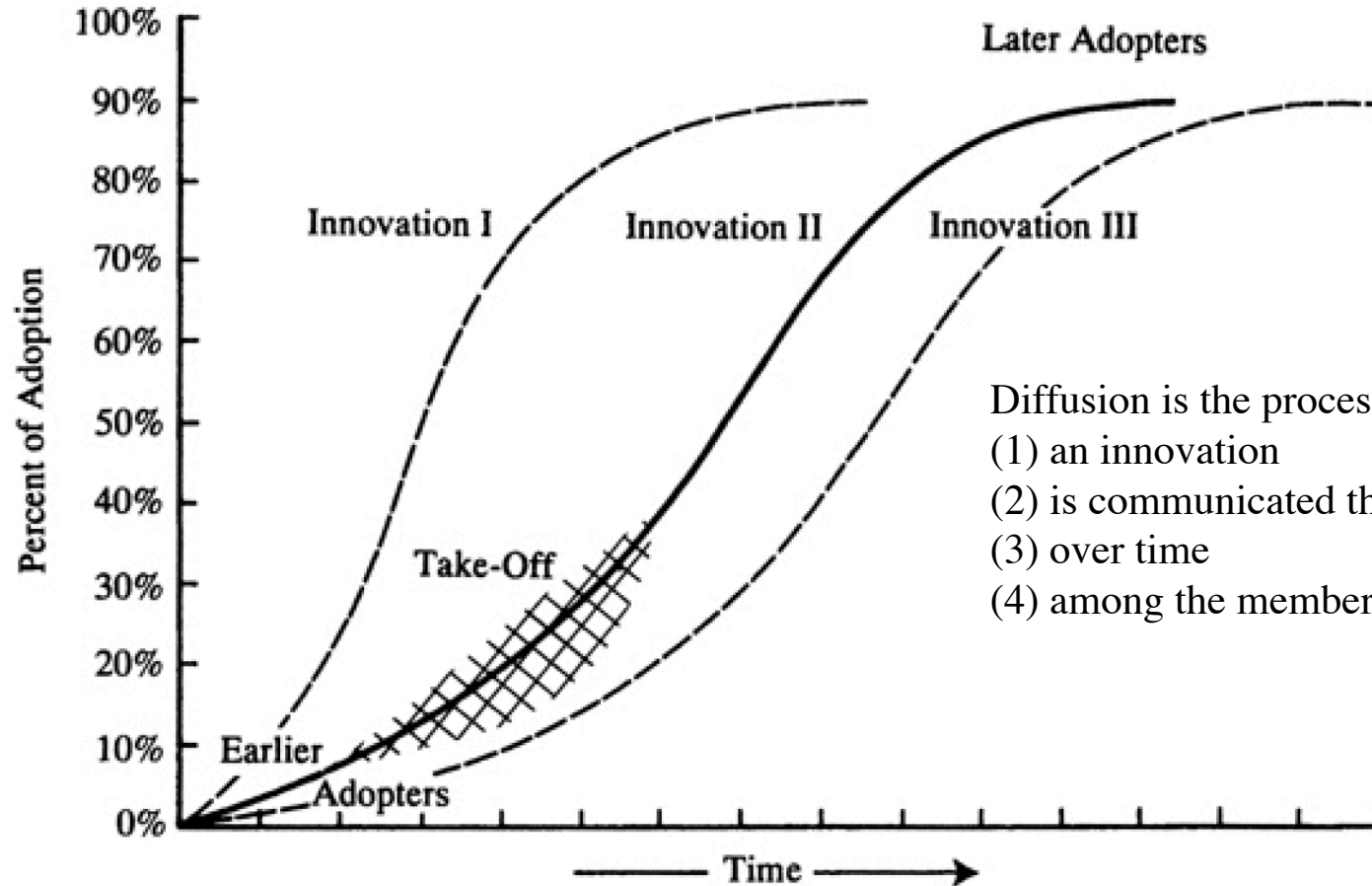
- V. Govindarajan & C. Trimble (2010) *The Other Side of Innovation: Solving the Execution Challenge*. Harvard Business Review Press.
- G. A. Moore (1995) *Inside the Tornado: Strategies for Developing, Leveraging, and Surviving Hypergrowth Markets*. Collins Business Essentials.
- E. M. Rogers (1995) *Diffusion of Innovations*. Fifth Edition, Free Press.



Diffusion of Innovations



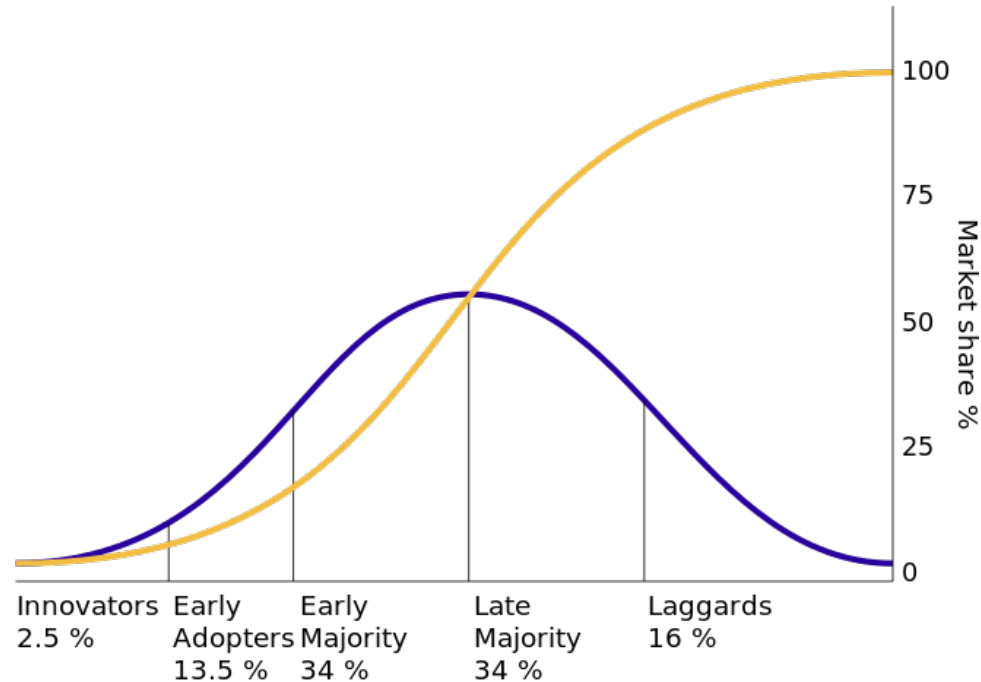
- Proposed general model of research/innovation diffusion
 - Popularised much later in Marketing by Frank Bass
- *Diffusion* is the process by which an innovation (i.e. new idea) is communicated through certain channels over time among the members of a social system.
- Tested in more than 6000 research studies and field tests
- In Diffusion of Innovations it is not people who change, but the innovations themselves (re-invention)



Diffusion is the process by which

- (1) an innovation
- (2) is communicated through certain channels
- (3) over time
- (4) among the members of a social system

Diffusion of Innovations



Diffusion of Innovations



- Five qualities determine between 49 and 87 percent of the variation in the adoption of new products.
 1. Relative advantage
 2. Compatibility with existing values and practices
 3. Simplicity and ease of use
 4. Trialability (degree innovation experimented with on a limited basis)
 5. Observable results

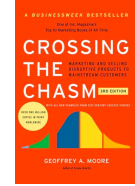
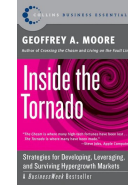
- Critical Mass occurs when the diffusion process becomes self-sustaining



Diffusion of Innovations

- Name a fast spreading technology? Early adopters? Any laggards? Reasons?
- What group are you generally in (fast adopter, laggard)?

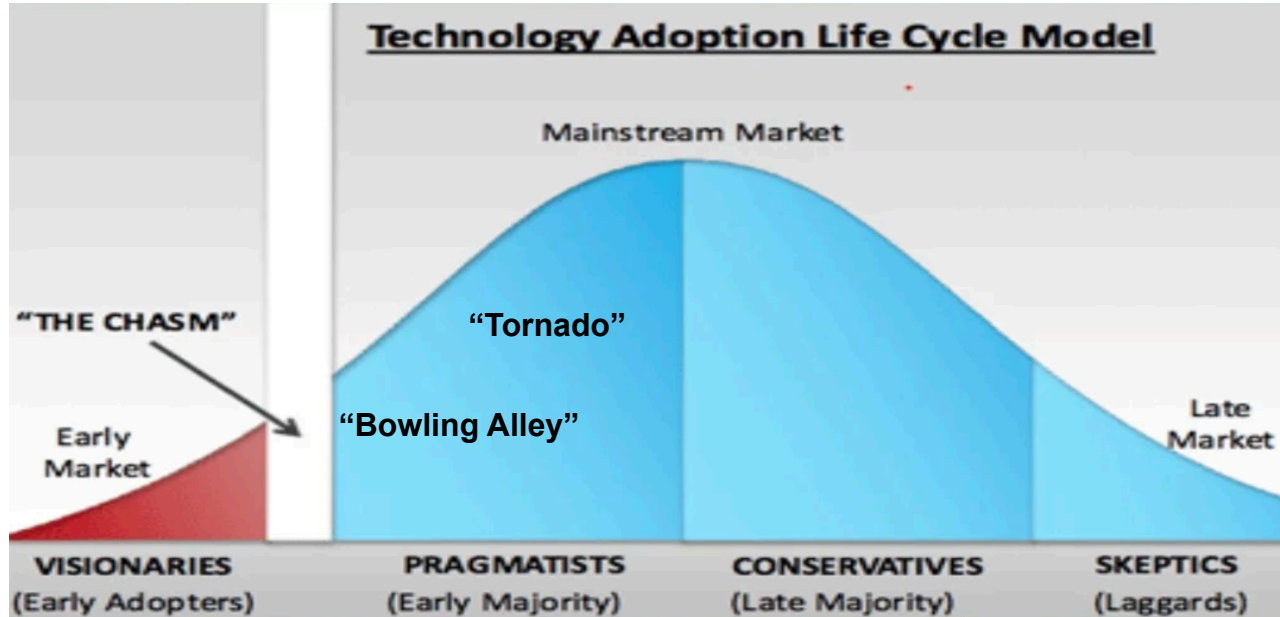
Inside the Tornado



- Diffusion of Technology General - Moore focuses on Technology Sector
- Proposes strategies to be successful in moving to next stage
 - Chasm make or break moment
 - Tornado when niche foothold and expand into main market
- Redefines, introduces terms (e.g. “mainstream market”)
 - Innovators = Technology Enthusiasts
 - Early Adopters = Visionaries
 - Early Majority = Pragmatists
 - Late Majority = Conservatives
 - Laggards = Skeptics

Inside the Tornado

Disruptive Technology
Mobile Phones
3d Printers
Social Networks
Cloud Computing



Inside the Tornado - Adoption Strategies

- Early Market: Go ahead of the herd for competitive advantage
 - Bespoke projects from the experts
 - Bowling Alley: Go ahead of the herd to address a painful problem
 - Proven applications from segment-focused customer
 - Tornado: Stick with the herd as it transitions to new infrastructure
 - Next-generation infrastructure from market leaders
 - Mainstream Market: Go after the herd for less disruption and a lower price
 - Evolutionary extensions from established customers
- No Budget

Debi Andrus's review of Inside the Tornado, The Journal of Marketing.



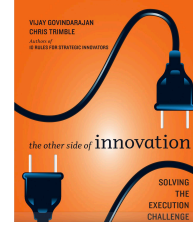
- Lack of evidence to support (cursory/personal experience)
- Ignores how innovation comes about in first place.
- Only applies to technology sector, if solid model should be validated in other sectors
- Can one model help practitioners improve their chances of success?
- How do professionals understand and use academic research?
- How can academic research truly assist professionals, and is that an important role?

Inside the Tornado

- What strategy does Moore suggest to cross the chasm?
- Do you agree more with Debi Andrus's or with Geoffrey Moore?



The Other Side of Innovation



- “Genius is 1 percent inspiration and 99 percent perspiration”
- Innovation **execution** is what this book is all about

Addressing the Fundamental Incompatibilities

- innovation = ideas
- innovation = ideas + execution
- innovation = ideas + motivation (e.g. Nucor bonuses, small)
- innovation = ideas + process (e.g. Deere and Company, similar)
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- innovation = ideas + leaders (setup for failure)
- innovation = ideas + leader + team + plan (organising/planning)

Addressing the Fundamental Incompatibilities

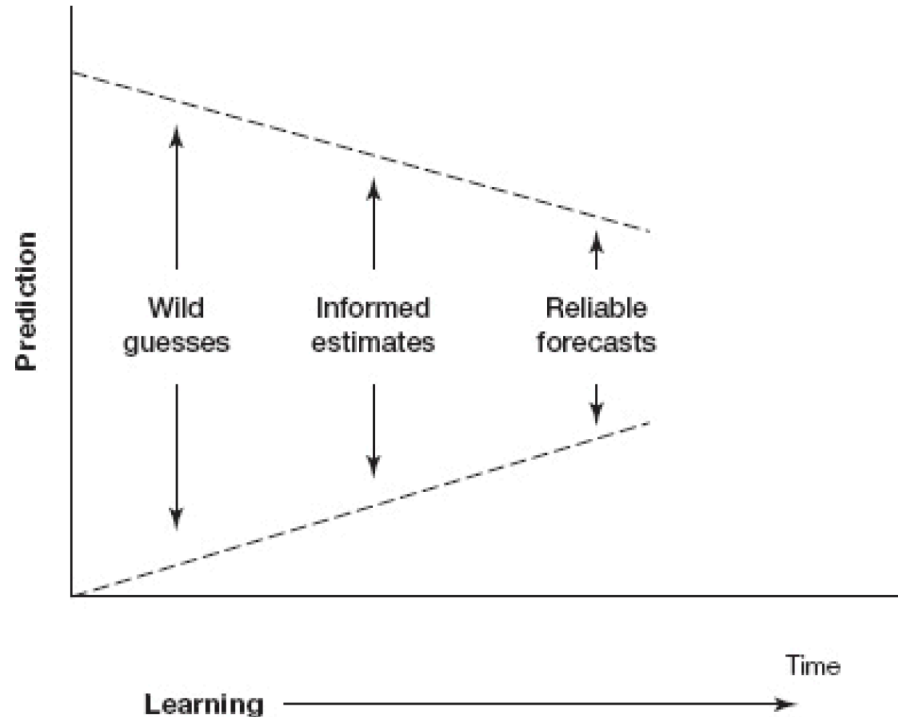
- Companies designed to be efficient (repeatable, predictable)
- Innovation non-routine, un-predictable
- Innovation and ongoing operations are always and inevitably in conflict.
- Because ongoing operations are *repeatable*, while innovation is *non-routine*, innovation leaders must think very differently about *organising*.
- Because ongoing operations are *predictable*, while innovation is *uncertain*, innovation leaders must think very differently about *planning*.

Addressing the Fundamental Incompatibilities

1. The real innovation challenge lies beyond the idea.
2. Performance Engine can innovate similar to past efforts.
3. Incompatibilities between innovation and ongoing operations.
Impossible for the Performance Engine to innovate on its own.
4. Innovation leaders (rebels) fighting the establishment. But one person against the bureaucratic octopus is bad bet.
5. Each innovation initiative requires a team with a custom organisational model and a project-specific plan.
6. Despite the inevitable tensions, innovation leaders must strive for a relationship of mutual respect with the Performance Engine.

Addressing the Fundamental Incompatibilities

- Best indicator that learning is taking place, predictions improve





The Other Side of Innovation

- According to Govindarajan, should every one in company be focused on innovation?
- Keep separated from core business (like skunkworks)?

Question/Discussion

- We started with: After you've come up with an innovation, what next? How do innovations spread and become popular?
- Assume you (and perhaps colleagues) are part of a large car manufacturing company and came up with a novel AI innovation (you should describe this in 2-3 sentences)
 - Explain how you would use Innovation Diffusion, Crossing the Chasm, and Other Side of Innovation to make it popular.

Identify Recent Innovation and Identify Groups

- 20 minutes to answer one of the posed questions.

- 10 peer review your neighbour's answer
Use the university's marking criteria (Level 7)



<https://www.bristol.ac.uk/academic-quality/assessment/regulations-and-code-of-practice-for-taught-programmes/marketing-criteria/>

10 discuss/explain/rebut with your neighbour